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PALEOMAGNETISM OF THE LEUCITE HILLS VOLCANIC FIELD, SOUTHWESTERN WYOMING

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Abstract. Paleomagnetic results have been determined for 41 sites, in lava flows and plug domes, in the 300 km^2 Leucite Hills volcanic field (1.1 ± .4 mybp), southwest Wyoming. Rock magnetic and petrographic studies show that magnetite is the important remanence carrier. Primary spinel and rhombohedral phases are both present. Analysis of VGP positions from 35 of these sites yields a paleomagnetic pole at 113.3°E, 87.4°N (α_{95} = 7.2°, δ = 23.0°) which, as expected, is coincident with the present day rotation axis. Generally, the ultrapotassic rocks of the Leucite Hills are magnetically stable; both polarities are recorded in the field along with a few directions which may be transitional. About 90% of the sites are reversely magnetized. Due to a lack in precision of the age determination for the Leucite Hills, the normal flows cannot be constrained to the Brunhes epoch or either the Cobb Mountain or Jaramillo events.

Introduction

The Leucite Hills volcanic field, southwest Wyoming, is a well exposed, largely unaltered assemblage of ultrapotassic lava flows, cinder and lava cones, dikes and plug domes (Figure 1). The volcanic field covers approximately 300 km^2; individual flows have areal extents on the order of 10^{-3} km^2 to 1 km^2. Flow thickness ranges up to about 40 m but flows thicker than 15 m are infrequent (Ogden, 1979). Volcanic activity in the Leucite Hills is relatively recent. An analyzed phlogopite crystal has been dated by the K-Ar method at 1.1 ± .4 mybp by McDowell (1971).

The Leucite Hills volcanic rocks are unusual. Chemically similar igneous rocks have only two other known occurrences, western Australia and southwestern Spain. A summary of the mineralogy and petrology of the Leucite Hills volcanics has been presented by Carmichael (1967). This reconnaissance has been followed by an extensive investigation by Ogden (1979). Because the rocks present in the Leucite Hills are not common a short description is of interest here. All igneous activity in the Leucite Hills is represented by three rock types: wyomingite, orendite and madupite. Of these three, madupite is known from no other place. The three lithologies were defined by Cross (1897) during one of the earliest studies of the Leucite Hills. Aphanitic groundmasses with phlogopite phenocrysts are common to all. Wyomingite and orendite both contain leucite, diopside, apatite and glass in the groundmass; orendite has sanidine in the groundmass also. Chemically, wyomingite and orendite are indistinguishable and they occur together, intermixed, at the hand sample scale (Ogden, personal communication). Madupite is mineralogically similar to the other two and is distinguished on a textural basis; it has poikiolitic phenocrysts of phlogopite enclosing diopside in a leucite, diopside, apatite groundmass.

Classical magnetostratigraphy is not likely to add to our understanding of the evolution of the Leucite Hills because volcanics from the various buttes and mesas rarely overlap or interfinger in space. Hence, even if opposing polarities were present at every sampling locality temporal overlapping and interfingering throughout the volcanic field could not be uniquely determined. In a situation such as this, opposite magnetic polarity can rule out a proposed correlation. However, we felt that this study might allow us to make a contribution to the general geology of the volcanic field by obtaining an estimate of the duration of volcanic activity. This character of information could be gained by two methods: 1) documenting the presence of both polarities of the magnetic field, thereby suggesting a minimum time span on the order of 10^3-10^4 years per transition, and 2) obtaining a measure of the angular dispersion of the paleomagnetic field. If the duration of volcanism were short compared to paleosecular variation (PSV) angular dispersion of VGP positions would be small. Similarly, if eruption sampled a longer time span a greater angular variance between remanence directions for various flows would be expected. In addition, the study would

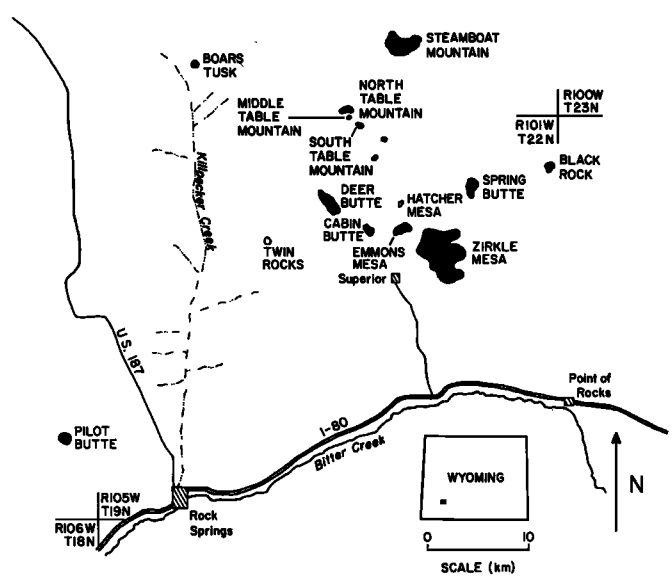


Fig. 1. Index map of the Leucite Hills.

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TABLE 1. Paleomagnetic data from the Leucite Hills volcanic field. N, number of samples; H_o, optimum demagnetization field; D, declination; I, inclination; α_{95}, radius of 95% confidence cone; δ, site dispersion; K, best estimate of precision parameter (α_{95}, K, from Fisher (1953)); Plat, VGP latitude, °N; Plong, VGP longitude, °E; #, rejected sites, see text for explanation.

Site	N	H_o	D	I	α_{95}	δ	K	Plat	Plong
Spring Butte 1	6	200	207.2	-61.5	8.7	9.5	60.7	69.9	-30.2
#Spring Butte 2	6	600	181.2	-58.2	26.8	27.9	7.2	86.9	53.6
Spring Butte 3	8	400	200.4	-60.0	4.3	5.8	168.3	74.7	-22.2
Spring Butte 4	7	400	214.7	-61.2	2.5	15.2	24.5	64.4	-31.7
S. Table Mt. 1	6	200	190.8	-49.9	9.6	10.5	49.5	75.9	29.6
S. Table Mt. 2	6	200	195.2	-48.3	7.4	8.1	83.1	72.3	22.0
S. Table Mt. 3	5	600	186.6	-52.5	6.3	6.0	148.6	79.8	38.0
S. Table Mt. 4	5	600	194.9	-53.1	16.4	15.3	22.6	75.7	11.3
S. Table Mt. 5	5	400	192.3	-52.3	5.5	5.2	194.0	76.8	19.8
Mid. Table Mt.	8	600	129.9	-81.4	7.9	10.7	50.6	51.0	-129.7
Steamboat Mt. 1	6	400	174.8	-58.3	14.8	16.0	21.6	85.1	126.2
Steamboat Mt. 2	6	400	169.0	-49.8	6.9	7.5	96.6	75.7	112.4
Steamboat Mt. 3	7	200	158.3	-67.4	10.6	13.0	33.2	72.9	-162.5
Steamboat Mt. 4	8	200	170.9	-69.1	8.4	11.4	44.2	77.7	-135.8
Hatcher Mesa	6	400	145.2	-46.8	14.6	15.8	22.1	58.6	146.2
Deer Butte 1	5	400	123.0	-66.6	16.9	15.7	21.6	50.3	-168.3
Deer Butte 2	7	400	173.6	-70.8	8.9	11.0	46.6	76.1	-124.4
Cabin Butte	6	800	166.7	-44.4	13.0	14.1	27.6	70.8	110.1
Emmons Mesa 1	6	600	223.1	-62.9	20.4	21.7	11.7	58.8	-38.5
Emmons Mesa 2	5	400	208.3	-62.6	6.8	6.5	126.5	69.3	-34.3
Emmons Mesa 3	5	600	195.0	-62.5	8.6	8.1	81.1	78.8	-34.3
Zirkel Mesa 1	5	400	198.4	-60.8	9.5	8.9	66.3	76.3	-25.1
Zirkel Mesa 2	5	400	188.5	-62.5	9.7	9.1	63.5	83.4	-39.3
#Zirkel Mesa 3	5	800	68.1	-82.4	40.2	34.4	4.6	34.9	-126.0
Zirkel Mesa 4	6	400	178.0	-69.6	19.1	20.4	13.2	78.4	-114.8
Black Rock	6	200	20.6	56.6	8.4	9.2	64.4	73.5	-8.9
#Boars Tusk 1	6	800	98.0	-44.8	33.5	33.7	5.0	22.9	-183.6
#Boars Tusk 2	6	600	147.4	-38.5	50.6	46.2	2.7	56.0	134.4
Boars Tusk 3	5	400	178.7	-48.4	10.8	10.1	51.6	77.4	76.1
Boars Tusk 4	4	600	153.6	-53.0	18.1	13.6	26.7	67.6	147.5
Matthews Hill	6	400	163.1	-51.0	11.5	12.5	35.1	73.1	129.0
Badger Teeth 1	6	400	159.6	-22.8	10.4	11.4	42.4	55.2	107.5
#Badger Teeth 2	5	600	140.2	-26.6	29.9	26.3	7.8	45.8	134.0
Badger Teeth 3	6	400	145.0	-16.5	23.3	24.5	9.2	44.6	123.6
Badger Teeth 4	5	400	154.3	-9.5	20.5	18.9	14.9	46.5	109.7
Badger Teeth 5	5	600	166.9	-30.4	8.2	7.7	88.1	62.2	98.6
Pilot Butte 1	6	200	339.4	66.3	11.6	12.7	34.1	73.9	-166.5
Pilot Butte 2	8	200	333.5	64.1	8.3	11.2	45.7	70.5	-178.0
Pilot Butte 3	5	200	347.1	48.9	8.8	8.3	77.0	74.2	116.3
#N. Table Mt. 1	6	600	173.9	-62.5	26.3	27.4	7.4	85.1	174.0
N. Table Mt. 2	10	400	235.0	-67.5	6.6	10.4	55.1	51.9	-51.1

also yield interesting results about Quaternary paleosecular variation and about the magnetic properties of these unusual rocks.

Sampling and Data Collection

During the summers of 1977 and 1979, samples oriented in situ were collected from 41 paleomagnetic sites in the Leucite Hills volcanic field. All three lithologies are represented in the sample population. Rock drills born of chain saws were used to collect from 5 to 10 samples at each site. Orientation of the cores was done using magnetic compasses; back azimuths were taken to insure against introduction of error from local magnetic anomalies.

Natural remanent magnetization (NRM) of specimens cut from the oriented samples was measured on a Schonstedt SSM-1 spinner magnetometer. About half the sites were subjected to stepwise alternating field demagnetization with peak fields of 50, 100, 200, 400, (and sometimes) 600, 800, and 1000 oersted. For the remaining sites, the 50 oersted level was omitted. The optimum demagnetization level for each site was chosen as the level which maximized the best estimate of the precision parameter K (Fisher, 1953). The alternating field demagnetization was performed, over a period of many months, with two different demagnetizing apparatuses: 1) a four axis tumbler system with inductrol field control and, 2) a Schonstedt GSD-1 specimen demagnetizer. When tested for consistency, both units gave equivalent results at the same de-

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magnetization level even after storage tests of several months. Most samples responded well to alternating field demagnetization tests. Essentially all samples used in the statistical analysis had attained a stable end point at the optimum demagnetization field. Several sites were randomly chosen for storage tests; invariably they returned to their previous directions after alternating field demagnetization at 200 to 400 oersteds.

The cutoff level for site reliability was arbitrarily established by excluding sites from statistical analysis which had 95% confidence cones with radius greater than 25°; this excluded six sites. Although said sites were deemed unreliable for inclusion in pole position or angular dispersion calculations, observation of their behavior during determination of magnetic stability showed that they are probably reliable indicators of polarity. All six of these sites were magnetized while the magnetic field was in a reversed state (see Table 1). In addition to the above discussed reliability criterion, individual samples that remained at more than twice the angular standard deviation from their respective site mean directions at the optimum demagnetization level were excluded from further statistical analysis; this criterion affected 13 samples.

No tectonic (tilt) corrections were necessary. The largest, discernable, post-volcanism fault offset is 1.5 m (Ogden, 1979). Field evidence indicates that the volcanics have remained as formed with no differential rotations between sites and minimal erosional decay of the volcanic field.

Magnetic Mineralogy

Investigation of the magnetic mineralogy of these rocks is important particularly in view of 1) their unusual overall mineralogy and 2) a curious result of an earlier study by Carmichael (1967). Carmichael found that there were no iron-titanium oxides in the wyomingites and orendites of the Leucite Hills. This is in direct conflict with our observations of: 1) mean NRM intensities of about 10^{-3} emu/cm^3, and 2) the stable magnetic behavior of most of our sites. This evidence suggested the presence of some iron-titanium oxides in the volcanics.

Initial attempts to determine the nature of the magnetic mineralogy by the Curie temperature technique were thwarted; even at fields as low as one to two kilogauss any ferrimagnetic effects were obscured by paramagnetic signals. However, a Curie temperature determination on a concentration of the magnetic phase from one sample revealed the presence of magnetite (Figure 2). Reflected light microscopy of polished thin sections from several representative samples also revealed the presence of iron-titanium oxides. Many of these grains were too small for accurate determination of composition, but primary phases of both spinel and rhombohedral structure are present. The paleomagnetic and thermomagnetic evidence mentioned above indicates that spinels close to magnetite in composition are the dominant remanence carriers.

The average volume susceptibility of the Leucite Hills volcanics is about 10^{-5} cgs. Assuming that magnetite is the dominant remanence

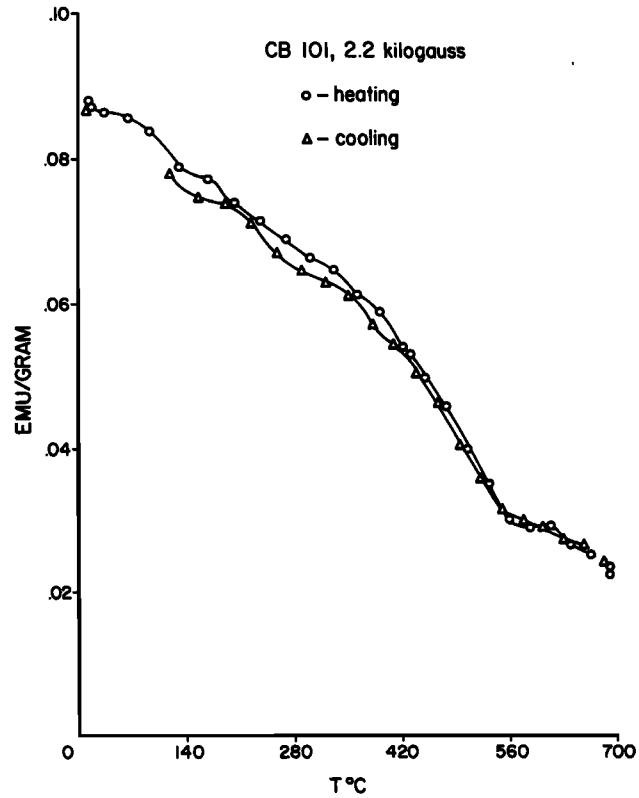


Fig. 2. Thermomagnetic analysis curve for a sample with concentrated magnetic fraction. The Curie temperature indicates magnetite.

carrier and using the relationship $\chi \simeq .2$ F of Stacey and Banerjee (1974), F, the volume fraction of magnetite, is on the order of $.5 \times 10^{-4}$ for these rocks. This estimate of small volume fraction is consistent with the noted difficulties of optical detection.

Results

Paleomagnetic data determined for the sites of this study are presented in Table 1. Of the 41 sites, all but 4 are reversely magnetized. Curiously, the sites with normal polarity are from 2 mesas, Pilot Butte and Black Rock, that represent the western and eastern limits of the volcanic field.

Virtual geomagnetic poles (VGP) were calculated for all sites (Table 1); their northern hemisphere positions are shown in Figure 3. The angular standard deviation (Wilson, 1959) for the group of 35 VGP is 23.0°. This term δ_t consists of 2 components: 1) the field dispersion, δ_f and 2) the within site dispersion, δ_w; $\delta_t^2 = \delta_f^2 + (\delta_w^2 / \bar{n})$ (McElhinny and Merrill, 1975) where δ_w is the average dispersion at each of N sites and $\bar{n}$ is the average number of samples per site. The field dispersion for this study is 22.4°. This value is larger than generally predicted by paleosecular variation models (Baag and Helsley, 1974). Several of the VGP depicted in Figure 5 are 35° to 40° or so from the population mean (longitude 113.3°E and latitude 87.4°N, $\alpha_{95} = 7.2°$) which is statistically coincident with the present rotation axis. VGPs at this angular distance from the

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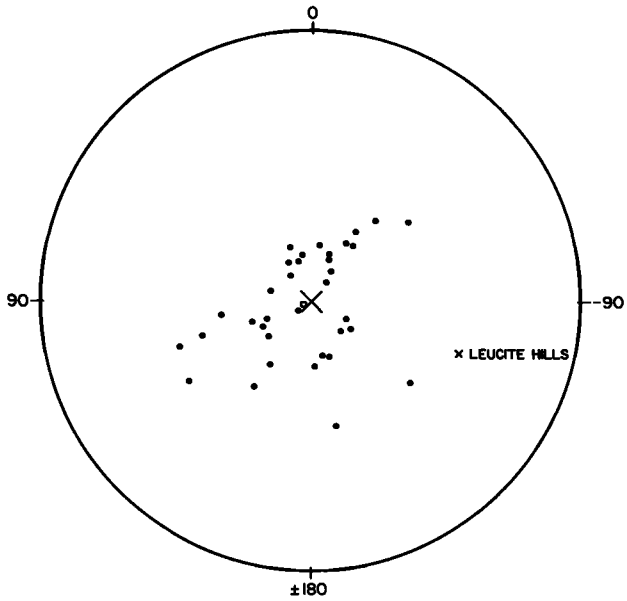


Fig. 3. Equal-area plot of 35 VGP considered in statistical analysis. The square represents the mean.

known rotation axis are quite likely to represent sites that have sampled a polarity transition, an excursion of the pole, or some other period of abnormal geomagnetic behavior. The presence of two antipodal polarities, combined with the greater than expected angular dispersion and the aberrant VGP positions, suggests that part of a polarity transition has been recorded during the evolution of Leucite Hills volcanism. However, the rocks do not give a very well documented picture of paleofield behavior during the event. There are no VGPs with colatitude greater than 45°. Although our sites are probably not all independent with respect to paleofield behavior, our sampling of the volcanic field has been comprehensive, and it is unlikely that further sampling would provide a greater number of independent measurements of paleofield behavior.

The observation that the geomagnetic field underwent a complete reversal (180°) during the evolution of the volcanic field shows that the minimum duration of volcanic activity was on the order of 10^3 - 10^4 years. That volcanism lasted longer than this is supported by the large angular dispersion recorded by the rocks. If the volcanic field evolved rapidly and uniformly during the transition a progression of north VGPs to the equator and back would be expected. If the volcanic field evolved rapidly just before or after the transition most of the VGPs would be tightly clustered with only a few positions marking the transition. The distribution of VGPs recorded by the volcanic field (Figure 3) is unlike either of these two cases. If we omit sites with VGP colatitudes greater than 35°, it is obvious that the remaining VGPs from sites with reversed polarity are the result of volcanic activity having had enough time to average the paleosecular variation to a constant, reasonable value. This time may be on the order of 10^5 to 10^6 years. These estimates of duration of volcanism are broad and tentative; it is not possible to set an absolute boundary on the duration, but the published radiometric age (1.1 ± .4 mybp) of McDowell (1971) provides an additional, compatible constraint. This range of .8 my spans the more recent half of the Matuyama reversed epoch, during which the geomagnetic field was predominantly in the reversed state. Farther back into the Matuyama reversed epoch normal polarity becomes more frequent. More recent than the Matuyama epoch is the Brunhes normal epoch. If time uniform volcanic activity persisted earlier or later than 1.1 ± .4 mybp then more frequent occurrences of normal polarity would be expected from the data. As it is, greater than 90% of the Leucite Hills sites were magnetized during a reversed state of the geomagnetic field. Hence, duration of volcanism was most likely confined to the most recent half of the Matuyama reversed epoch. The available radiometric date is not precise enough to constrain the time of magnetization of sites with normal polarity (Black Rock and Pilot Butte) to the early Brunhes epoch, the Jaramillo event or the Cobb Mountain polarity event observed by Mankinen et al. (1978).

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References

Baag, C., and Helsley, C.E., Geomagnetic secular variation model E., J. Geophys. Res., 79, 4918-4922, 1974.

Carmichael, I. S. E., The mineralogy and petrology of the volcanic rocks from the Leucite Hills, Wyoming, Contr. Mineral. Petr., 15, 24-66, 1967.

Cross, W., Igneous rocks of the Leucite Hills and Pilot Butte, Wyoming, Am. J. Sci. 4th ser., 4, 115-141, 1897.

Fisher, R. A., Dispersion on a sphere, Proc. Roy. Soc. Lond., Ser. A, 217, 295-305, 1953.

Mankinen, E. A., Donnelly, J., and Grommé, C. S., Geomagnetic polarity event recorded at 1.1 mybp on Cobb Mountain, Clear Lake volcanic field, California, Geology, 6, 653-656, 1978.

McDowell, F., K-Ar ages of igneous rocks from the western United States, Isochron/West, 2, 1-16, 1971.

McElhinny, M. W., and Merrill, R. T., Geomagnetic secular variation over the past 5 m.y., Rev. Geophys. Space Phys., 13, 687-708, 1975.

Ogden, P. O., The geology, major element geochemistry, and petrogenesis of the Leucite Hills volcanic rocks, Wyoming, Ph.D. Thesis, University of Wyoming, 135 pp., 1979.

Stacey, F. D., and Banerjee, S. K., The Physical Principles of Rock Magnetism, (Elsevier Scientific Publishing Co., Amersterdam), 195 pp., 1974.

Wilson, R. L., Remanent magnetism of the late secondary and early Tertiary British rocks, Philos. Mag., 4, 750-755, 1959.

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